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SOLAR WATER HEATER CONFIGURATION TUBE SERPENTINE WITH VARIATION OF SINGLE GLASS, DOUBLE GLASS AND DOUBLE GLASS VACUUM CONVECTORS

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ABSTRACT

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The purpose of this study was to determine the performance of a solar water heater after adding variations to the collector cover, namely double glass and vacuum double glass. The method used in this research is field observation by conducting direct tests on flat plate solar collectors with a serpentine tube configuration. Measurements and data collection are carried out per hour from 09.00-16.00 WIB. The addition of glass as a collector cover to 2 cover glasses is enough to reduce heat loss and slightly increase the efficiency. The lowest heat loss occurred in the first hour of the double glass vacuum cover variation, namely $U_t = 0.96 \text{ W/m}^2 \text{ } ^\circ\text{C}$ (48.24%), side heat loss $U_{se} = 0.15 \text{ W/m}^2 \text{ } ^\circ\text{C}$ (7.54%), bottom heat loss $U_b = 0.88 \text{ W/m}^2 \text{ } ^\circ\text{C}$ (44.22%) and total heat energy loss $U_L = 1.99 \text{ W/m}^2 \text{ } ^\circ\text{C}$. The highest efficiency occurred in the double glass variation, which was 40.59%. From the calculations that have been done, it can be said that the use of a collector with a double glass cover and a double vacuum glass is slightly more effective than only a single glass.

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Keywords: Solar water heater, flat plate collector, efficiency, heat loss

1. INTRODUCTION

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The potential for solar energy in Indonesia is very abundant because it is a tropical country (Sidopekso 2011). The sun is a renewable source of solar energy that is abundant, easy to obtain, environmentally friendly and easy to use. The sun has a source of solar energy that can be utilized for everyday life. One of the benefits of solar power is heating water. Hot water can be produced in many ways, including using a fire to boil water, then using solar power to heat water.

There are many ways to harness this abundant solar energy, including solar cells and thermal solar (Artiningrum and Havianto 2019). Technology with solar cells is relatively efficient and clean, but requires quite expensive equipment. Meanwhile, solar

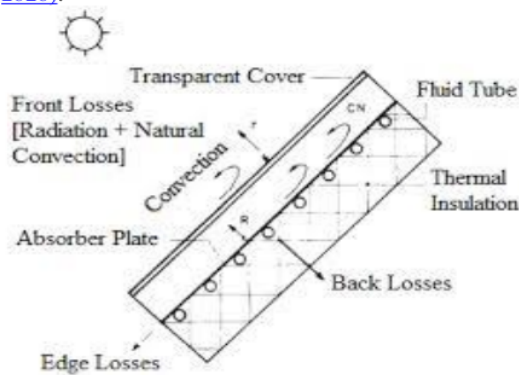
thermal technology collects solar radiation in the form of heat. This method generally does not require complicated tools and is relatively easy to do. Globally, the use of solar thermal is still far more than solar cells (Purba and A.Sianturi 2022).

Flat plate solar collectors are one of the most widely used types of collectors because they are simpler than other types of collectors. The flat plate solar collector will absorb the sun's radiant energy and turn it into heat to heat the water in the pipes, so that the water temperature increases and circulation occurs due to differences in water density (Thermosiphon) (Hayati, Ginting, and Tambunan 2019). The purpose of this research is to reduce the amount of losses to the environment and obtain a more optimal temperature.

2. LITERATURE REVIEW

2.1 Solar energy

Solar energy is a renewable energy source which is the result of nuclear fusion reactions in the sun's core. Arriving at the surface of the earth, solar energy will be absorbed (51%), reflected, returned to space (30%) and into the atmosphere clouds (19%). Indonesia, which is located in the tropics, has a sizable advantage, namely receiving sun rays throughout the year (Yunianto 2020). This energy can be harnessed using a range of technologies such as solar heating, solar photovoltaic, solar heating electricity, solar architecture, and artificial photosynthesis. Solar energy technologies are generally categorized into two groups, namely active and passive technologies. This grouping depends on the process of absorption, conversion and distribution of solar energy (Zayyinun and Widyartono 2020).



Picture 1. Flat plate collector

2.2 Kalor

Heat is defined as a form of energy that can be transferred from one system to another as a result of a temperature difference, the transfer of heat energy is always from a higher temperature to a lower temperature, and heat transfer stops when the two media reach the same temperature (Cengel 2004). Heat is usually expressed in terms of temperature, the unit of heat in international units is the joule, another unit of heat is the calorie. 1 calorie is defined as the amount of heat needed to heat 1 kg of water by 1°C. 1 calorie = 4.2 joules and 1 joule = 0.24 calories, the size of the heat needed by an object (substance) depends on 3 factors, namely:

1. Mass of Substance
2. Type of substance (specific heat)
3. Temperature Changes.

2.3 Types of Heat Transfer

There are three types of displacement: conduction, convection, and radiation.

Heat transfer by **conduction** is the transfer of heat from a substance without the movement of the particles of that substance. The ability of objects to conduct heat is divided into three, namely, conductors, insulators, and semiconductors.

Heat transfer by **convection** is the transfer of energy between a solid and a moving liquid or gas, and it is a combination of conduction and fluid motion. The greater the heat transfer convection, the faster the fluid motion will be, and if it is not moving this condition is called pure conduction.

Radiation is energy emitted by matter in the form of electromagnetic waves (photons) in the form of changes in the electron configuration of atoms or molecules, radiation heat transfer can still occur in the absence of other media, unlike conduction and convection (Cengel 2004).

2.4. Definition and function of the serpentine tube type solar water heater

A serpentine tube type solar water heater is a tool that can heat water with the help of solar energy (Rusmaryadi, Veronika, and Purwanto 2018). In its use, this tool does not use electricity and fuel, so this tool is very environmentally friendly and energy efficient. This tool works by utilizing the sun's heat energy to heat water inside the solar collector.

2.5. The working principle of a solar heater

Solar water heater is a tool to convert cold water into hot water by involving solar heat. The working principle of this tool uses the thermosiphon phenomenon. Thermosiphon occurs because of the difference in the density of water, if the pipe inside the solar collector gets heat from the sun, then the density of water will decrease due to the difference in density between hot water and cold water, then there is a clockwise circulation, warm water from the solar collector transferred to a storage tank and cold water is pushed towards the solar collector (Sidopekso 2011).

2.6. Types of solar heaters

A solar water heater is a device used to convert sunlight into a form of heat energy for various purposes, such as heating water (Deceased and Beckman 2013).

Solar collectors have various forms, one of which is a prism that can receive sunlight from various directions of the sun. The advantage of a prismatic solar

collector is that it has the ability to receive solar radiation energy from various directions from the sun (Philip Kristanto and James Laeyadi 2000).

The flat plate solar collector is an elongated collector with the desired angle of inclination to capture solar energy (Sudrajat and Santosa 2014). The process of using it is arguably easier and simpler. Flat plate collectors have large heat sinks which can cause large amounts of heat loss as well.

Vacuum tube collectors have the ability to absorb solar radiation energy from a free angle without the need to concentrate on the incoming sunlight, these tubes are also superior to flat glass collectors (Harun et al. 2020).

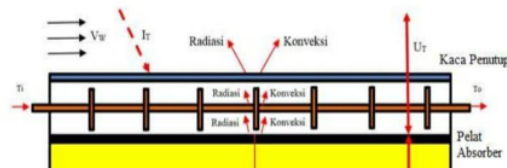
The insulator is used to retain heat which will leave the collector to the environment (Mahardhika et al. 2019). Types of insulators that are often used in solar water heaters:

Polystyrene is a polymer composed of the monomer styrene, a liquid hydrocarbon made from petroleum. The polystyrene thermal conductivity figure is 0.08 W/(m.K) .

Styrofoam is an ethylene-based synthetic polymer made from monomers from petroleum cracking. Styrofoam or foam is still included in the plastic group, generally Styrofoam is white and looks clean, the shape is also simple and light. Styrofoam is produced from a mixture of 90-95% polystyrene and 5-10% gas. The average thermal conductivity figure is 0.095 W/(m.K) (Hariady and Fauzie 2014).

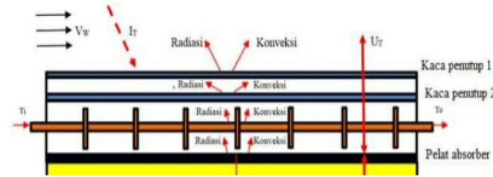
The collector cover glass functions as a successor to solar radiation and prevents heat from escaping at the top of the collector to the environment (Susanto and Irawan 2017).

1. Type one cover glass is a common type of solar collector. This type has better efficiency than a collector without a glass cover, because this type of collector has a single layer of glass which functions to keep heat out so that the heat inside the collector is maintained.



Picture 2. One cover glass collector

2. The two cover glass type has a sealed air gap between the two panes of glass acting as an insulator keeping heat out of the solar collector. So that the heat contained in the collector will be maintained with the help of the air gap which can increase the efficiency of the solar collector.



Picture 3. Two cover glass collector

3. This type is designed to produce higher heat energy than the previous two types of solar collectors. This tool has a high heat transfer efficiency, but the heat loss factor is relatively low. This is because the trapped air is conditioned in a vacuum so that it can reduce the rate of heat loss by convection from the surface to the environment.

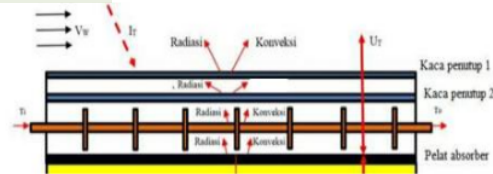


Figure 4. Double glass cover and vacuum collector

3. RESEARCH METHODOLOGY

- Literature Study Method

A study to get a comprehensive picture of what other people have done and how people have done it, then how different the research will be.

- Field Observation Method

This method first makes observations of the object to be studied, this observation is carried out to obtain information about human behavior as in reality, we can get this observation with a more open picture with the aim of knowing more and exploring the problems that occur in life and are related to using research processes.

The testing procedure in this study is as follows:

1. The collector is placed at an angle of 23.5°
2. The measurement time starts at 09.00-16.00 WIB.
3. Data collection time per hour

This research was conducted in February 2023 at the Energy Conversion Laboratory, Mechanical Engineering Study Program, Faculty of Engineering, Tridianti University, Palembang.

4. RESULTS AND DISCUSSION

This calculation provides an example of a complete calculation of the double glass test on only one data, namely February 15, 2023 at 09.00.

$$T_{s1} = 31^\circ\text{C}$$

$$T_{k1} = 46^\circ\text{C}$$

$$T_s = 50 \text{ minute}$$

$$v = 1.7 \text{ km/h}$$

$$T_{s2} = 32^\circ\text{C}$$

$$T_{k2} = 50^\circ\text{C}$$

$$G = 253 \text{ W/m}^2$$

$$T_a = 27^\circ\text{C}$$

First, calculate **Insolation** (solar radiation on a horizontal surface in 1 hour), namely:

$$\begin{aligned} G &= 253 \text{ W/m}^2 = 253 \text{ (J/s)/m}^2 \\ &= 253 \text{ (J/3600 hour)/m}^2 \\ &= 910.800 \text{ J/(hour.m}^2) \\ I &= 0.910 \text{ MJ/(hour.m}^2) \end{aligned}$$

Where: I is the emitted radiation energy for 1 hour. And, at steady state, the output energy of the solar collector is the difference between the solar radiation absorbed by the collector and the loss of thermal energy from the solar collector (Deceased and Beckman 2013), namely:

$$Q_u = A_c [S - U_L (T_{pm} - T_a)]$$

Where :

Q_u = Energy output (J/s)

A_c = Collector area (m^2) ($= 1.2\text{m} \times 0.8\text{m} = 0.96 \text{ m}^2$)

S = Solar radiation absorbed per unit area (W/m^2)

U_L = Coefficient of loss or total loss of solar collector ($\text{W/m}^2 \text{ } ^\circ\text{C}$)

T_a = The temperature around the collector is 27°C

T_{pm} = Average plate temperature = $(46^\circ\text{C} + 50^\circ\text{C}) / 2 = 48^\circ\text{C}$

The known quantities include; number of cover glass ($N = 2$), plate to cover glass distance (17 mm), glass emittance ($\epsilon_g = 0.94$), plate emittance ($\epsilon_p = 0.17$), collector slope ($b = 23.50$), bottom insulation thickness ($L = 4\text{cm}$), conductivity of styrofoam insulation ($k = 0.035 \text{ W/m } ^\circ\text{C}$), length x width x height of collector box $1.2\text{m} \times 0.8\text{m} \times 20\text{cm}$, side insulation thickness 4cm, Stefan-Boltzmann constant ($\sigma = 5.66697 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$), latitude position of Palembang city ($2^\circ 59'$ or rounded up to 3°), solar radiation constant ($G_{sc} = 1367 \text{ W/m}^2$), diffuse reflectance ($r_g = 0.6$), KL of glass/glass ($KL = 0.037$), and plate absorptivity (normal incidents) = 0.6.

Heat loss from the top of the collector box is dominant due to wind. The heat transfer coefficient from the wind can be calculated by the equation (Deceased and Beckman 2013):

$$h_w = 2.8 + 3.0 \cdot v = 2.8 + 3 \cdot (1.7 \cdot 1000/3600) = 4.2 \text{ W/m}^2$$

Where:

h_w = wind heat transfer coefficient ($\text{W/m}^2 \text{ } ^\circ\text{C}$)

v = wind speed (km/hour or m/s).

Then, the amount needed is calculated one by one. For example, from (Deceased and Beckman 2013) obtained:

$$\begin{aligned} f &= (1 + 0.089 h_w - 0.1166 h_w \epsilon_p) (1 + 0.07866 N) \\ f &= (1 + 0.089 (4.2 \text{ W/m}^2 \text{ } ^\circ\text{C}) - 0.1166 (4.2 \text{ W/m}^2 \text{ } ^\circ\text{C}) (0.17)) (1 + 0.07866 (2)) = 1.495 \end{aligned}$$

Then the C scale:

$$C = 520 \cdot (1 - 0.00005 I \cdot \beta^2)$$

Where: This formula is used when $0^\circ < \beta < 70^\circ$, Jika $70^\circ < \beta < 90^\circ$, use $\beta = 70^\circ$, so that

$$C = 520 (1 - 0.00005 (23.5)^2) = 505.4$$

Likewise with the magnitude of e

$$e = 0.430 (1 - 100/T_{pm})$$

$$e = 0.430 (1 - 100/321) = 0.296$$

The upper heat loss coefficient can be calculated using the equation (Deceased and Beckman 2013) :

$$U_t = \left[\frac{N}{\frac{c}{T_{pm}} \left(\frac{T_{pm} - T_a}{N + f} \right)^e + \frac{1}{h_{wind}}} \right]^{-1} + \frac{\sigma (T_{pm} - T_a) (T_{pm}^2 - T_a^2)}{\frac{1}{\epsilon_p + 0.0059 N h_w} + \frac{2N + f - 1 + 0.133 \epsilon_p}{\epsilon_g} - N}$$

$$U_t = \left[\frac{1}{\frac{505.4}{321} \left(\frac{321 - 300}{1 + 1.495} \right)^{0.296} + \frac{1}{4.2}} \right]^{-1} + \frac{5.67 \cdot 10^{-8} (321 - 300) (321^2 - 300^2)}{\frac{1}{0.17 + 0.0059 (2) (4.2)} + \frac{2 \cdot 2 + 1.495 - 1 + 0.133 (0.17)}{0.94} - 2}$$

$$U_t = 1.02 \text{ W/m}^2 \text{ } ^\circ\text{C}$$

The heat loss coefficient on the side of the collector box can be calculated by the equation (Deceased and Beckman 2013)

$$U_e = (U_{samping} \cdot A_{edge=sampling}) / A_c$$

where if the above equation is solved one by one its parts, will be obtained:

$$U_{samping} = k/L$$

Where :

k = thermal conductivity of the insulation (Styrofoam) = 0.035 W/m K

L = side insulation thickness ($= 4\text{cm} = 0.04\text{m}$)

$$\text{Jadi } U_{samping} = \frac{0.035 \text{ W/m}^2 \text{ } ^\circ\text{C}}{0.04 \text{ m}} = 0.0014 \text{ W/m}^2 \text{ } ^\circ\text{C}$$

Furthermore :

$$A_{edge=sampling} = (1.2 \text{ m} + 0.8 \text{ m} + 1.2 \text{ m} + 0.8 \text{ m}) \cdot 0.17 \text{ m}$$

$$A_{edge=sampling} = 4 \text{ m} \cdot 0.17 \text{ m} = 0.68 \text{ m}^2$$

also counted

$$A_c = 1.2 \text{ m} \cdot 0.8 \text{ m} = 0.96 \text{ m}^2$$

Loss of heat lost on the side

$$U_e = \frac{0.0014 \frac{\text{W}}{\text{m}^2} \cdot 0.68 \text{ m}^2}{0.96 \text{ m}^2} = 0.15 \text{ W/m}^2 \text{ } ^\circ\text{C}$$

The heat loss coefficient on the underside of the collector box can be calculated by the equation (Deceased and Beckman 2013) :

$$U_b = \frac{1}{R_4} = \frac{k}{L}$$

$$U_b = \frac{0.035 \text{ W/m}^2 \text{ } ^\circ\text{C}}{0.04 \text{ m}} = 0.88 \text{ W/m}^2 \text{ } ^\circ\text{C}$$

If the three types of losses are added together, a total collector loss coefficient will be obtained (Deceased and Beckman 2013), namely:

$$U_L = U_i + U_b + U_e$$

$$U_L = 1.72 \text{ W/m}^2\text{°C} + 0.88 \text{ W/m}^2\text{°C} + 0.15 \text{ W/m}^2\text{°C} = 2.74 \text{ W/m}^2\text{°C}$$

The total radiation on an inclined surface can be calculated using the equation (Deceased and Beckman 2013):

$$I_T = (I_b + I_d \cdot A_i) R_b + I_d (1-A_i) \left(\frac{1+\cos\beta}{2} \right) \left(1 + f \cdot \sin^3 \left(\frac{\beta}{2} \right) \right) + I \cdot \rho_g \left(\frac{1-\cos\beta}{2} \right)$$

However, there are still many quantities that are not known and need to be calculated first, namely:

- The value of n is obtained from table 1.6.1 (Deceased and Beckman 2013), namely $n = 31+i$ (February 15), so $n = 46$
- The δ value is obtained from the same table as above: $\delta = -14.6^\circ$
- The value of ϕ is the position of Palembang, namely: $2^\circ 59 \text{ LS}$, so $\phi = 3$
- The ω_s value obtained by the formula:
 $\cos \omega_s = -\tan \phi \cdot \tan \delta = -\tan(3) \cdot \tan(-14.6) = 89.29$
 Values ω_{s1} , ω_{s2} and ω , where: $\omega_{s1} = -45^\circ$ (data first at 09.00, where in 1 hour the sun shifts 15°), $\omega_{s2} = -30^\circ$ (first data at 10.00), and $\omega = (-45^\circ + (-30^\circ)) / 2 = -37.5^\circ$ (as the average value -flat).

- The value of I_o is obtained from:

$$I_o = \frac{12 \cdot 3600}{\pi} G_{sc} \left(1 + 0.003 \cos \frac{360n}{365} \right) \left(\cos \phi \cdot \cos \delta (\sin \omega_2 - \sin \omega_1) + \frac{\pi(\omega_2 - \omega_1)}{180} \sin \phi \cdot \sin \delta \right)$$

$$I_o = 3.818.178,13 \text{ J/m}^2$$

- Value of R_b obtained from:

$$R_b = \frac{\cos(\phi + \beta) \cdot \cos \delta \cdot \cos \omega + \sin(\phi + \beta) \cdot \sin \delta}{\cos \phi \cdot \cos \delta \cdot \cos \omega + \sin \phi \cdot \sin \delta}$$

$$R_b = 0.80$$

- K_T value is obtained from:

$$K_T = \frac{I}{I_o} = \frac{972.000}{3.594.829,28} = 0.243$$

- I_d/I value from (Deceased and Beckman 2013):

$$\frac{I_d}{I} = 0.9511 - 0.1604 K_T + 4.388 K_T^2 - 16.638 K_T^3 + 12.336 K_T^4 = 0.975$$

- I_d values are obtained from:

$$I_d = 0.766 \cdot I = 0.766 \cdot 910.800 = 888.455 \text{ J/m}^2$$

- I_b values are obtained from:

$$I_b = (1-0.966) \cdot 910.800 = 22.344,6 \text{ J/m}^2$$

- A_i values are obtained from:

$$A_i = \frac{I_b}{I_o} = \frac{22.344,6}{3.745.930,08} = 0.006$$

- f values are obtained from:

$$f = \sqrt{\frac{I_b}{I}} = \sqrt{\frac{22.344,6}{910.800}} = 0.157$$

If all the calculated values and quantities are included in the equation, the total radiation on the inclined surface will be obtained:

$$I_T = (I_b + I_d \cdot A_i) R_b + I_d (1-A_i) \left(\frac{1+\cos\beta}{2} \right) \left(1 + f \cdot \sin^3 \left(\frac{\beta}{2} \right) \right) + I \cdot \rho_g \left(\frac{1-\cos\beta}{2} \right)$$

$$I_T = 892.297,7 \text{ J/m}^2$$

Furthermore, the absorbed radiation can be calculated using the equation (Deceased and Beckman 2013):

$$S = I_b \cdot R_b (\tau\alpha)_b + I_d (\tau\alpha)_b \left(\frac{1+\cos\beta}{2} \right) + \rho_g \cdot I (\tau\alpha)_g \left(\frac{1-\cos\beta}{2} \right)$$

However, there are still many quantities that are not known and need to be calculated first, namely:

- Beam radiation

$$\theta = \frac{46 \left(\frac{\text{derajat}}{\text{gerakan matahari dari iUkes}} \right) - 35 \left(\frac{\text{selisih 15 Feb ke 22 mar}}{182 \left(\frac{\text{hari}}{\text{gerakan matahari dari iUkes}} \right)} \right)}{1} = 8.85$$

From Figure 4.11.1 (Deceased and Beckman 2013), for $\theta = 8.85^\circ$, we get $\frac{\alpha}{\alpha_n} = 0.996$ and from Figure 5.3.1

(Deceased and Beckman 2013) obtained $\tau = 0.70$ so that: $(\tau\alpha)_b = 1.01 \cdot 0.62 \cdot 0.996 \cdot 0.60 = 0.423$

- For diffuse radiation, we get $\theta = 57.5^\circ$ from Figure 5.4.1 (Deceased and Beckman 2013), from Figure 4.11.1 we get $\frac{\alpha}{\alpha_n} = 0.94$. Then from Figure 5.3.1 we get $\tau = 0.72$, so: $(\tau\alpha)_d = 1.01 \cdot 0.72 \cdot 0.94 \cdot 0.60 = 0.410$
- For beam radiation, we get $\theta = 78.5^\circ$ from Figure 5.4.1 (Deceased and Beckman 2013), from Figure 4.11.1 we get $\frac{\alpha}{\alpha_n} = 0.72$. Then from Figure 5.3.1 we get $\tau = 0.22$, so: $(\tau\alpha)_d = 1.01 \cdot 0.22 \cdot 0.7 \cdot 0.60 = 0.096$

If all values and quantities that have been calculated are entered into the equation, you will get:

$$S = I_b \cdot R_b (\tau\alpha)_b + I_d (\tau\alpha)_b \left(\frac{1+\cos\beta}{2} \right) + \rho_g \cdot I (\tau\alpha)_g \left(\frac{1-\cos\beta}{2} \right)$$

$$S = 357.939,2 \text{ J/m}^2$$

From all the results of the calculations that have been done, the calculation of the solar collector output energy is as follows:

$$Q_u = A [S - U_L(T_{pm} - T_a)]$$

$$Q_u = 0.96 \text{ m} \cdot [357.939,2 \text{ J/m}^2 - 2.74 \text{ W/m}^2\text{°C} \cdot (50^\circ - 27)] = 343.580,5 \text{ J/s}$$

And the efficiency of the solar collector can be calculated as follows:

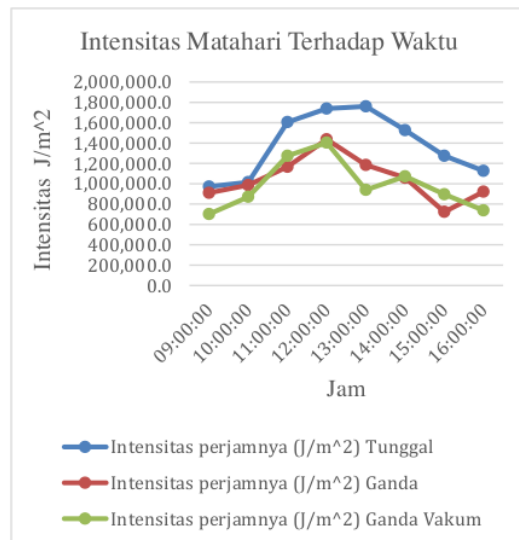
$$\eta = \frac{Q_u}{I_T A_c}$$

$$\eta = \frac{343.580,5}{892.297,7 \cdot 0.96} \cdot 100\%$$

$$\eta = 40.11\%$$

The results of measuring the intensity of solar radiation on an inclined plane with a collector tilt angle of 23.5°.

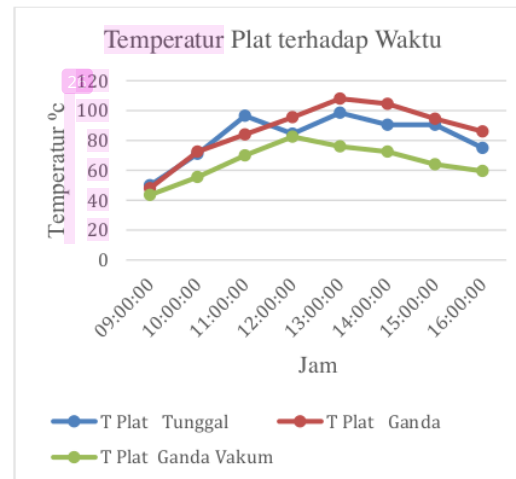
JAM	Intensitas perjamnya (J/m ²)		
	Tunggal	Ganda	Ganda Vakum
09:00:00	972,000.0	910,800.0	702,000.0
10:00:00	1,015,200.0	986,400.0	871,200.0
11:00:00	1,605,600.0	1,166,400.0	1,274,400.0
12:00:00	1,738,800.0	1,436,400.0	1,404,000.0
13:00:00	1,760,400.0	1,184,400.0	939,600.0
14:00:00	1,526,400.0	1,058,400.0	1,072,800.0
15:00:00	1,274,400.0	723,600.0	896,400.0
16:00:00	1,126,800.0	921,600.0	738,000.0



Based on the graph, data collection starts at 09.00 then will increase and will reach its peak between 11.00 and 13.00 WIB. After passing through these hours, the intensity of the sun will slowly decrease until late afternoon.

The results of collecting data on single glass collector plate temperature, double glass, and double glass vacuum.

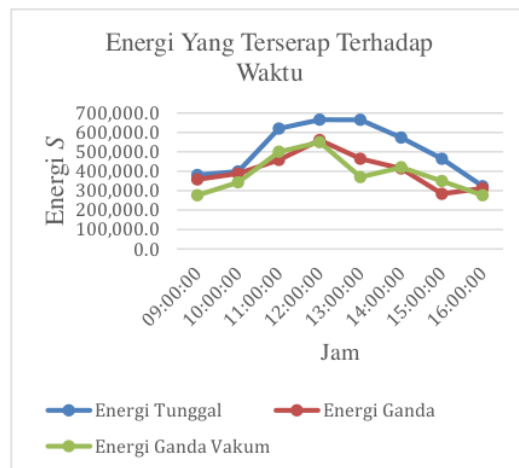
JAM	Temperatur Plat		
	Tunggal	Ganda	Ganda Vakum
09:00:00	50	48.0	43.5
10:00:00	71	72.5	55.5
11:00:00	96.5	84.0	70
12:00:00	84.5	95.5	82.5
13:00:00	98.5	108.0	76
14:00:00	90.5	104.5	72.5
15:00:00	90.5	94.5	64
16:00:00	75	86.0	59.5



The graph above shows an increase in temperature graphs every hour, this is in line with the increasing intensity of the sun. The temperature of the double glass collector is higher than the other two types because it has 2 cover glasses so that the reflection of sunlight from the collector will be reflected back in the two glasses not directly into the environment so that it can increase the temperature of the collector.

Data on the average energy absorbed by solar collectors with single glass, double glass and vacuum double glass covers.

JAM	Energi		
	Tunggal	Ganda	Ganda Vakum
09:00:00	380,953.2	357,939.2	276,352.1
10:00:00	399,104.8	388,138.9	343,078.1
11:00:00	620,040.1	458,504.2	500,468.7
12:00:00	665,699.7	560,921.8	549,707.8
13:00:00	664,966.3	464,571.0	369,984.0
14:00:00	573,146.5	414,296.4	420,394.3
15:00:00	463,935.7	283,876.4	349,424.5
16:00:00	322,995.7	312,177.5	276,204.5

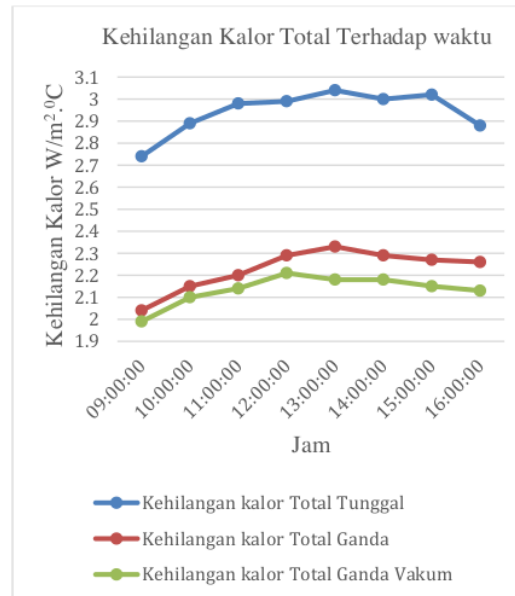


23

In the graph above it can be seen that the absorbed energy always increases every hour, this is because the high intensity of the sun absorbed by the collector will provide great energy to the absorber plate which is used to heat water.

The results of calculating the total heat loss in solar collectors with single glass cover, double glass, and vacuum double glass can be seen in the following table and graph.

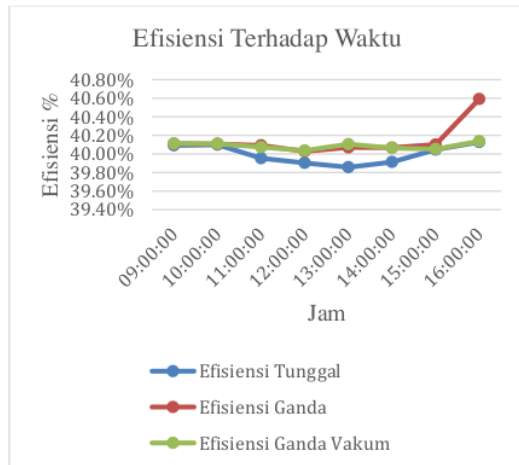
JAM	Kehilangan kalor Total		
	Tunggal	Ganda	Ganda Vakum
09:00:00	2.74	2.04	1.99
10:00:00	2.89	2.15	2.1
11:00:00	2.98	2.2	2.14
12:00:00	2.99	2.29	2.21
13:00:00	3.04	2.33	2.18
14:00:00	3	2.29	2.18
15:00:00	3.02	2.27	2.15
16:00:00	2.88	2.26	2.13



Based on the graph, the lowest total heat loss occurred in the test using a vacuum double glass, this occurred because there was no air between the two cover glasses so that convection did not occur and could reduce heat loss to the environment.

The results of efficiency calculations for solar collectors with single glass, double glass, and vacuum double glass covers can be seen in the following table and graph.

jam	Efisiensi		
	Tunggal	Ganda	Ganda Vakum
09:00:00	40.09%	40.11%	40.11%
10:00:00	40.10%	40.11%	40.11%
11:00:00	39.95%	40.09%	40.07%
12:00:00	39.90%	40.03%	40.04%
13:00:00	39.86%	40.07%	40.11%
14:00:00	39.91%	40.07%	40.06%
15:00:00	40.05%	40.10%	40.05%
16:00:00	40.13%	40.59%	40.14%



Based on the graph, it can be said that the double glass collector has slightly greater and better efficiency than the other two types. The results of this calculation depend on the intensity of the sun, the temperature of the water entering and leaving the serpentine pipe, and wind speed.

5. CONCLUSION

Based on solar collector testing with single glass cover, double glass, and vacuum double glass, it can be concluded as follows:

1. The solar collector does not work optimally due to erratic weather conditions such as cloudy.
2. The addition of glass as a collector cover to 2 cover glasses is enough to reduce heat loss and slightly increase efficiency. The lowest heat loss occurred in the first hour of the double glass vacuum cover variation, namely $U_t = 0.96 \text{ W/m}^2 \text{ } ^\circ\text{C}$ (48.24%), side heat loss $U_e = 0.15 \text{ W/m}^2 \text{ } ^\circ\text{C}$ (7.54%), bottom heat loss $U_b = 0.88 \text{ W/m}^2 \text{ } ^\circ\text{C}$ (44.22%) and total heat energy loss $U_L = 1.99 \text{ W/m}^2 \text{ } ^\circ\text{C}$. The highest efficiency occurred in the double glass variation, which was 40.59%.
3. From the calculations that have been done, it can be said that the use of a collector with a double glass cover and a double glass vacuum is slightly more effective than only a single glass.

Based on the tests and discussions carried out, the suggestions for further research are to extend or increase the bends in the water flow serpentine pipes and to collect data when the weather conditions are very good in order to get more optimal result.

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